Exercise 6.1

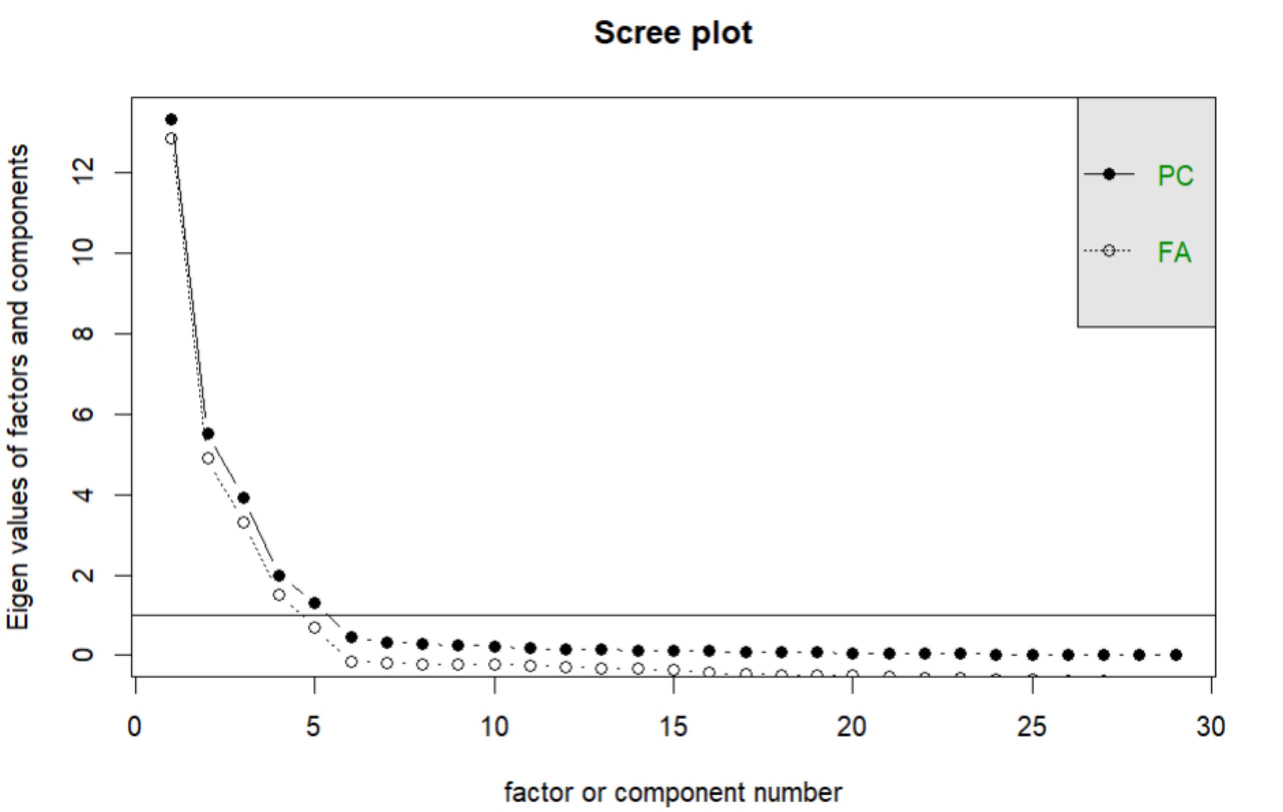
Case: Human resources and quality management

(1) The problem: The company want to clarify how quality of interactions, specifically leader-member exchange (LMX) and team-member exchange (TMX), affects employee self-efficacy (EFF) and job satisfaction (SAT), thereby ultimately lead to better quality management (TQM).

FA: used to discover factors for the quality management and relationships between factors and variables in the survey questions, and to test the model consistency with the data

SEM: forming qualitative models, thereby validating the model to interpolate the relationship between factors and their effects on quality management

1. (EFA) Firstly, create a scree plot of the data by the function scree(dat), collecting the plot below:



Based on the plot, there are five factors whose eigenvalues are over 1, so it can be said that **in this case, the appropriate number of factors is 5.**

Running KMO(dat), collecting the KMO test result:

Overall MSA = 0.82

q1	q2	q3	q4	q5	q6	q7	q8	q9	q10	q11	q12	q13	q14	q15	q16	q17
0.76	0.87	0.74	0.80	0.81	0.85	0.90	0.69	0.80	0.71	0.88	0.83	0.80	0.77	0.71	0.78	0.85

q18	q19	q20	q21	q22	r1	r2	r3	r4	r5	r6	r7
0.84	0.86	0.79	0.85	0.83	0.95	0.84	0.81	0.83	0.92	0.85	0.93

The value of KMO test is over 0.5, so the data is sufficient.

Running dev(cov(dat)), which calculates the determinant test, collects the result: 6.433468e-23. The number is too small, so it is not likely to cause multicollinearity.

Because of the assumption of the dependence between factors, the oblique rotation is used for EFA, so the function used in the EFA is fit.efa <- efa(dat, nfactors = 5, rotation = 'oblimin') (oblimin is used because of its precision).

Running summary(fit.efa, fit.measures = TRUE), collect the standardized loadings: (* = significant at 1% level)

	f1	f2	f3	f4	f5
q1	0.863*				
q2	0.924*				
q3	0.951*				
q4	0.940*				
q5	0.987*				
q6	0.915*				
q7		0.796*	. .		
q8		0.932*			
q9		0.938*			
q10		0.687*	. .		
q11		0.841*			
q12		0.909*	.		
q13	.		0.756*		
q14			0.943*		
q15			0.910*		
q16			0.919*		
q17			0.930*	.	
q18				0.846*	.
q19	.			0.838*	.

q20	0.932*	
q21	0.978*	
q22	0.873*	
r1		0.936*
r2		0.949*
r3		0.960*
r4		0.959*
r5		0.926*
r6		0.980*
r7		0.938*

With values over 0.6, it can be safe to say that these values show the association between the factors and variables, specifically:

- f1 is associated with variables q1 -> q6: LMX

- f2 is associated with variables q7 -> q12: TMX

- f3 is associated with variables q13 -> q17: EFF

- f4 is associated with variables q18 -> q22: SAT

- f5 is associated with variables r1 -> r7: TQM

The EFA fosters the theoretical measurement structure of the social exchange model.

2. (CFA) Based on the result of EFA and the assumed variable, establish the CFA model as follows:

model.cfa <- '

LMX =~ q1 + q2 + q3 + q4 + q5 + q6

TMX =~ q7 + q8 + q9 + q10 + q11 + q12

EFF =~ q13 + q14 + q15 + q16 + q17

SAT =~ q18 + q19 + q20 + q21 + q22

TQM =~ r1 + r2 + r3 + r4 + r5 + r6 + r7’

Running the summary(model.cfa, fit.measures = TRUE), collects some important measures:

- CFI: 0.948

- TLI: 0.942

- RMSEA: 0.084

RMSEA is the root mean squared error of approximation. It is considered smaller than 0.05 to indicate better match between the model and data, but in this case it reaches 0.084. However, the CFI and TLI are relatively high. CFI is Comparative Fit Index and Tucker-Lewis Index. If they are over 0.9, it means the model is better and ensures parsimony. Therefore, **the model ensures quality of fit**.

Running some functions for measures of construct reliability and convergent and discriminant validity:

comprelSEM(fit.cfa):

LMX TMX EFF SAT TQM

0.981 0.962 0.971 0.972 0.992

AVE(fit.cfa):

LMX TMX EFF SAT TQM

0.900 0.821 0.878 0.876 0.956

Htmt(model.cfa, dat):

	LMX	TMX	EFF	SAT	TQM
LMX	1.000				
TMX	0.114	1.000			
EFF	0.271	0.679	1.000		
SAT	0.371	0.310	0.181	1.000	
TQM	0.394	0.379	0.375	0.682	1.000

The comprelSEM measures the reliability of the model, which is better with higher values, at least over 0.7. The AVE measures convergent validity, which is better with values higher than 0.5. The HTMT measures the discriminant validity, which is better with values lower than 0.9. Based on the results above, it can be safe to say that **the measurement models via CFA are in acceptable ranges regarding construct reliability and convergent and discriminant validity, and do not need any more appropriate actions.**

(SEM)

1. The first model, where all latent variables have a direct effect only on TQM, is demonstrated below:

```
model.sem1 <- '

LMX =~ q1 + q2 + q3 + q4 + q5 + q6

TMX =~ q7 + q8 + q9 + q10 + q11 + q12

EFF =~ q13 + q14 + q15 + q16 + q17
```

SAT =~ q18 + q19 + q20 + q21 + q22

TQM =~ r1 + r2 + r3 + r4 + r5 + r6 + r7

TQM ~ LMX + TMX + EFF + SAT’

Fit the model: fit.sem1 <- sem(model.sem1, dat)

Assessing measures of construct reliability and convergent and discriminant validity, collecting the results below:

compRelSEM(fit.sem1): LMX TMX EFF SAT TQM

0.981 0.962 0.971 0.972 0.992

AVE(fit.sem1): LMX TMX EFF SAT TQM

0.900 0.821 0.878 0.876 0.956

htmt(model.sem1, dat):

	LMX	TMX	EFF	SAT	TQM
LMX	1.000				
TMX	0.114	1.000			
EFF	0.271	0.679	1.000		
SAT	0.371	0.310	0.181	1.000	
TQM	0.394	0.379	0.375	0.682	1.000

Running summary(fit.sem1, fit.measures = TRUE), collecting the indexes below:

- CFI: 0.948
- TLI: 0.942
- AIC: 2502.236
- BIC: 2644.651

The second model where LMX and TMX have direct effects on TQM, EFF and SAT, and EFF and SAT have a direct effect on TQM:

model.sem2 <- '

LMX =~ q1 + q2 + q3 + q4 + q5 + q6

TMX =~ q7 + q8 + q9 + q10 + q11 + q12

EFF =~ q13 + q14 + q15 + q16 + q17

SAT =~ q18 + q19 + q20 + q21 + q22

TQM =~ r1 + r2 + r3 + r4 + r5 + r6 + r7

EFF ~ LMX + TMX

SAT ~ LMX + TMX

TQM ~ LMX + TMX + EFF + SAT'

Fit the model: fit.sem2 <- sem(model.sem2, dat)

Assessing measures of construct reliability and convergent and discriminant validity, collecting the results below:

compRelSEM(fit.sem2): LMX TMX EFF SAT TQM

0.981 0.962 0.971 0.972 1.010

AVE(fit.sem2): LMX TMX EFF SAT TQM

0.900 0.821 0.878 0.876 0.972

htmt(model.sem2, dat):

	LMX	TMX	EFF	SAT	TQM
LMX	1.000				
TMX	0.114	1.000			
EFF	0.271	0.679	1.000		
SAT	0.371	0.310	0.181	1.000	
TQM	0.394	0.379	0.375	0.682	1.000

Running summary(fit.sem2, fit.measures = TRUE), collecting the indexes below:

- CFI: 0.948
- TLI: 0.942
- AIC: 2500.922
- BIC: 2641.243

All the measures regarding CFI, TLI, comprelSEM, AVE and htmt from the two models are in acceptable ranges, so no reason to eliminate any one for validation.

AIC is Akaiki Information Criterion, BIC is Bayesian Information Criterion. These numbers measure the likelihood of the data with the given model, and the pair of models with lower indexes would be preferred. From the above results, we can see that **the AIC and BIC of model 2 are lower than those of model 1, so model 2 is the better model.**

2. Using Model 2, to consider the net effect of TMX, creating a model that validates the mediated effect:

model.net <- '

LMX =~ q1 + q2 + q3 + q4 + q5 + q6

TMX =~ q7 + q8 + q9 + q10 + q11 + q12

EFF =~ q13 + q14 + q15 + q16 + q17

SAT =~ q18 + q19 + q20 + q21 + q22

TQM =~ r1 + r2 + r3 + r4 + r5 + r6 + r7

EFF ~ LMX + a1*TMX

SAT ~ LMX + a2*TMX

TQM ~ b1*EFF + b2*SAT + LMX + b3*TMX

b_ind := a1*b1+a2*b2

b_net := b_ind + b3'

b_ind is to calculate the indirect effect, while the b_net is to calculate the total net effect of TMX

Fit the model: fit.net <- sem(model.net, dat)

After running summary(fit.net, fit.measures = TRUE), collect the results below:

- CFI: 0.948

- TLI: 0.942

-> The model ensures the quality of fit.

Defined Parameters:

<table>
<tr><th></th><th>Estimate</th><th>Std.Err</th><th>z-value</th><th>P(>|z|)</th></tr>
<tr><td>b_ind</td><td>0.591</td><td>0.215</td><td>2.755</td><td>0.006</td></tr>
<tr><td>b_net</td><td>0.604</td><td>0.218</td><td>2.776</td><td>0.006</td></tr>
</table>

The CFI and TLI are over 0.9, so the model is qualitatively fit. Based on the results, it is noticeable that the indirect effect of TMX on TQM, which EFF and SAT mediated, is statistically positive (**$\beta = 0.591$**, $p = 0.006 < 0.05$, so the null hypothesis of b_ind being zero can be rejected). Also, the total net effect of TMX on TQM is considerable (**$\beta = 0.604$**, $p = 0.006 < 0.05$, so the null hypothesis of b_net being zero can be rejected). These points out that the team-member exchange affects the quality management directly and by driving the employees' self-efficacy and job satisfaction.

(3) Managerial implications:

- The EFA and CFA prove the assumption that **the quality management strongly follows the social exchange model, which is robustly driven by relationships and interactions in workplaces.**

- The model 2 being better **confirms that self-efficacy and job satisfaction mediate the relationship between quality of interaction and total quality management, which means that the feeling of confidence and contentment plays an important role for employees to contribute to the quality improvement.**

- The mediated model of TMX **indicates that stronger teamwork relationships strongly enhance the total quality management both directly and indirectly, emphasizing the essence of colleagues' support and collaboration.**

=> ***From the managers' perspective, it is essential not to overlook the latent effect of employees' experience with interactions with leadership and teammates on the quality management. The company should encourage leadership training and especially team cooperation, thereby enhancing the work psychology of employees, ultimately driving direct and indirect improvement of the business quality.***